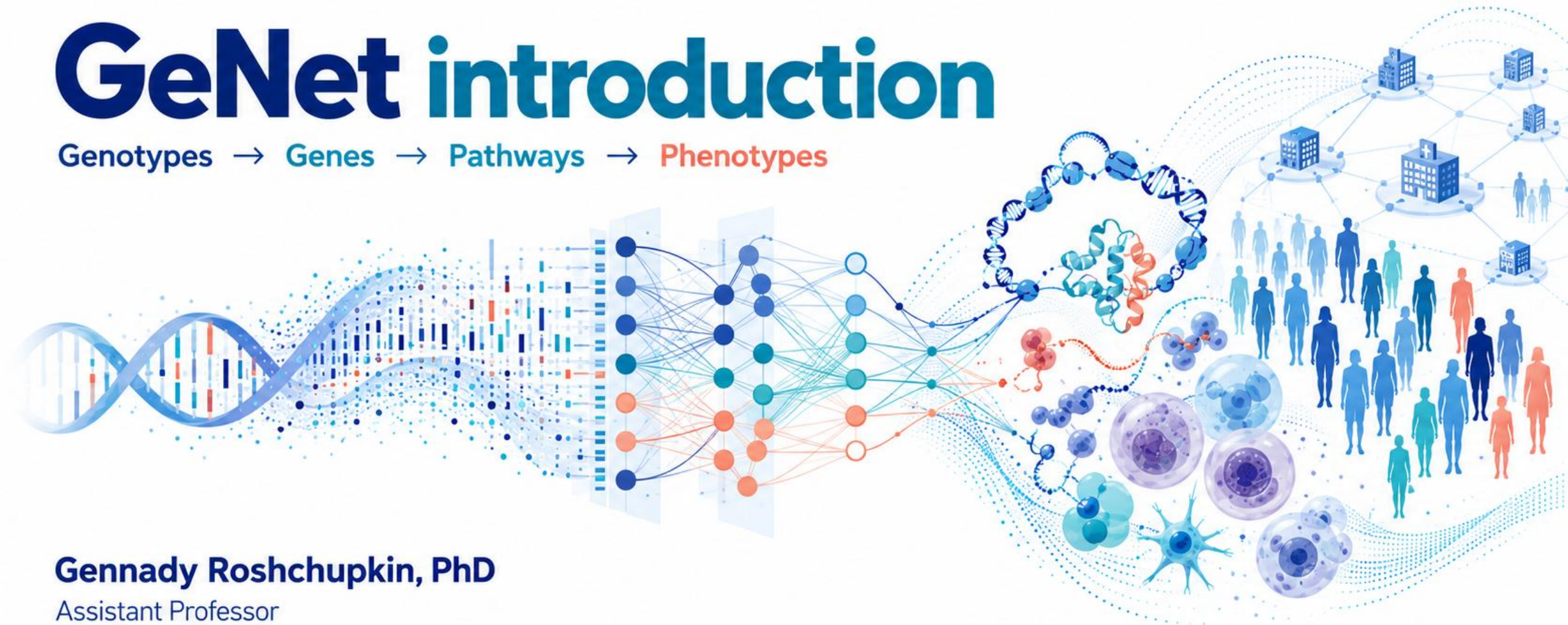


GeNet introduction

Genotypes → Genes → Pathways → Phenotypes



Gennady Roshchupkin, PhD

Assistant Professor

PI, Computational Population Biology

Erasmus MC, Rotterdam

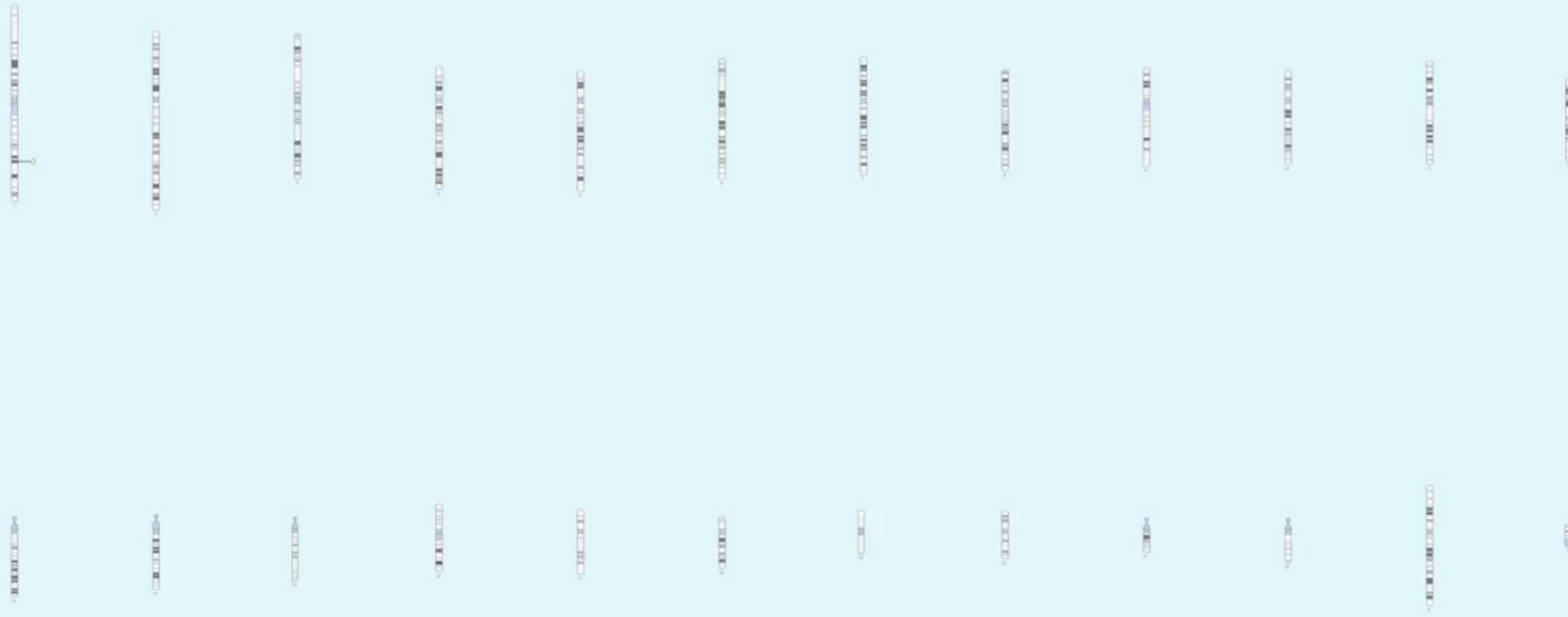
Epidemiology • Radiology & Nuclear Medicine

Email: g.roshchupkin@erasmusmc.nl

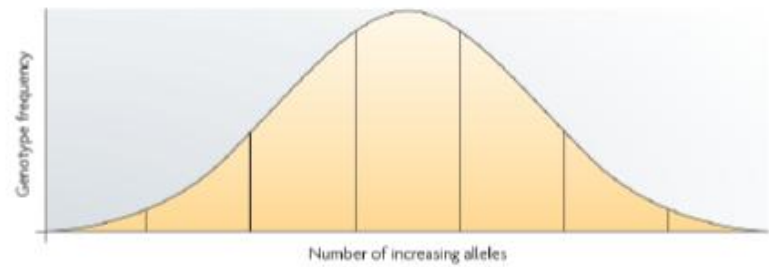
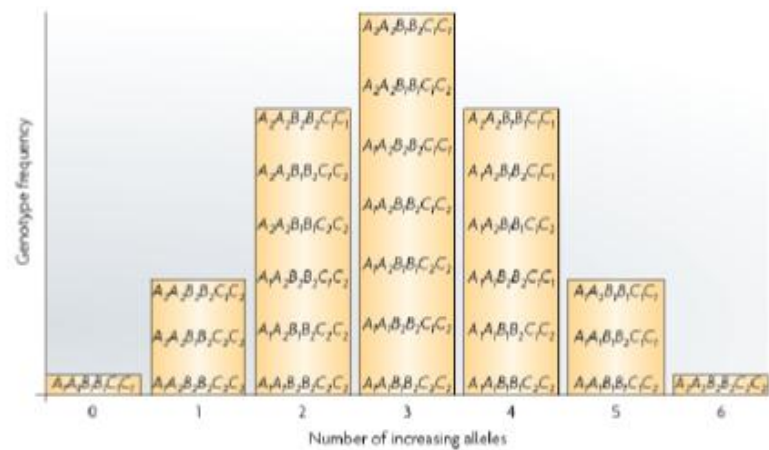
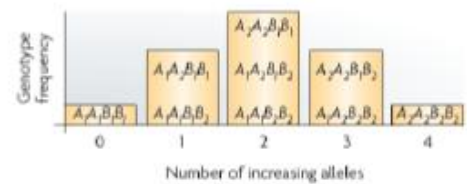
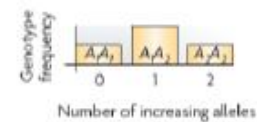
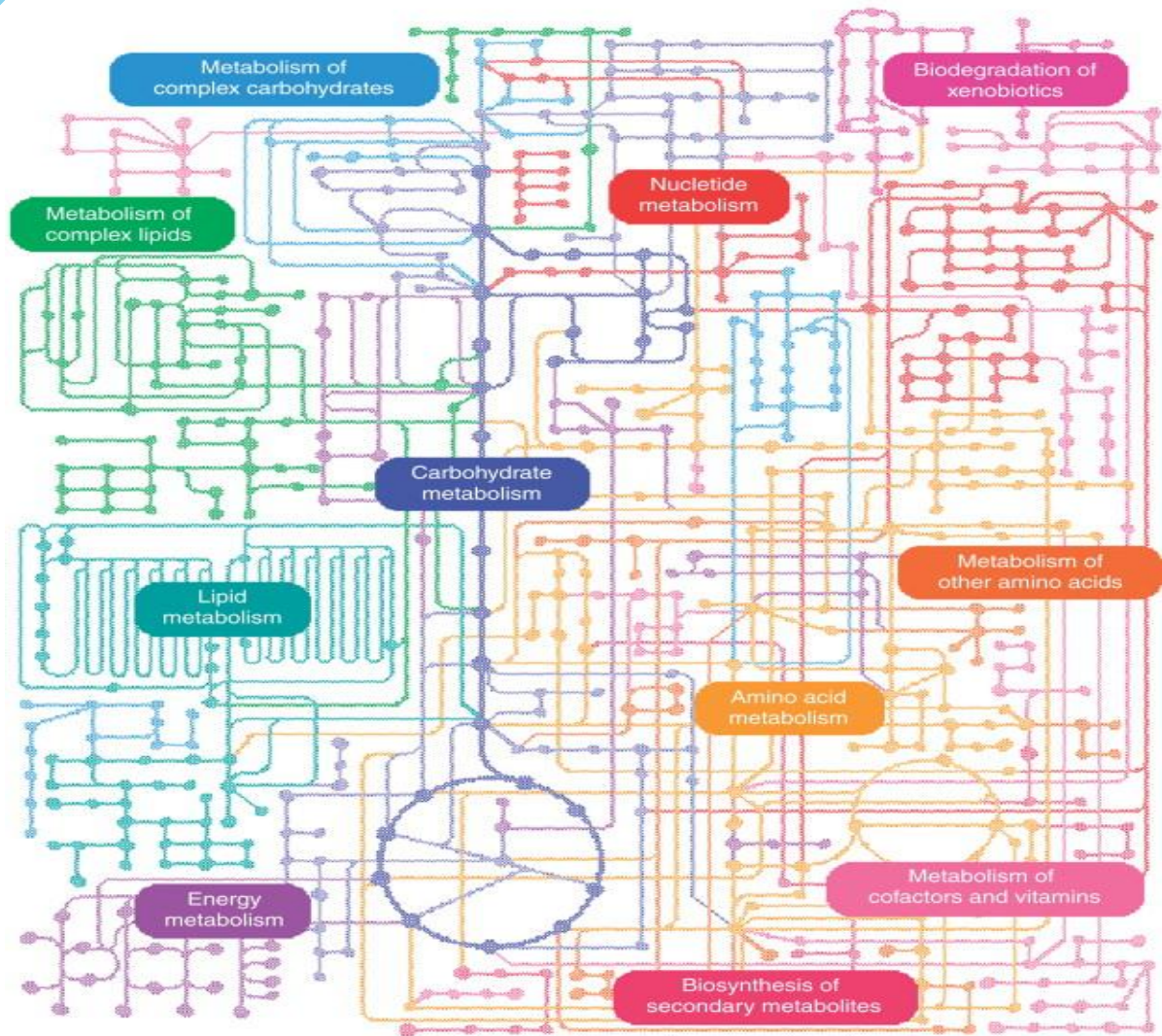
Twitter: [@Roshchupkin](https://twitter.com/Roshchupkin)

Web: www.roshchupkin.org / www.bigr.nl

2006 Jan

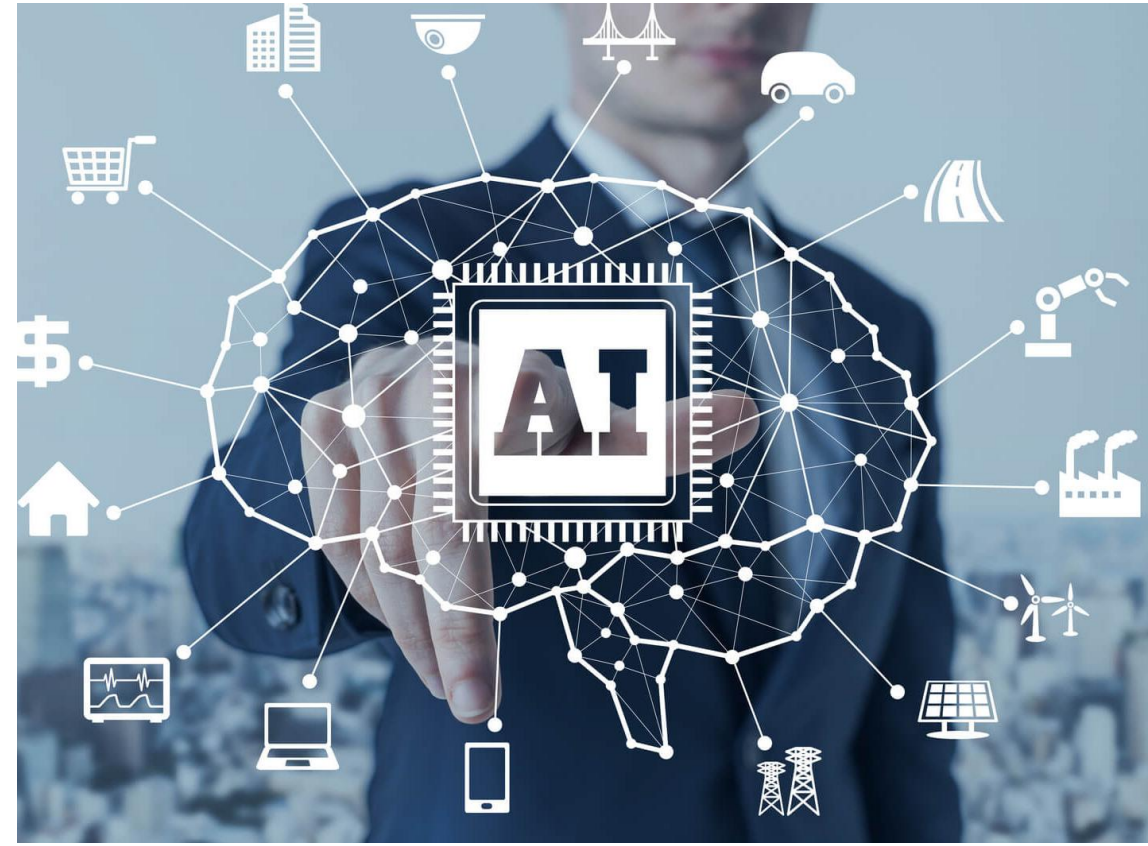


www.ebi.ac.uk/gwas

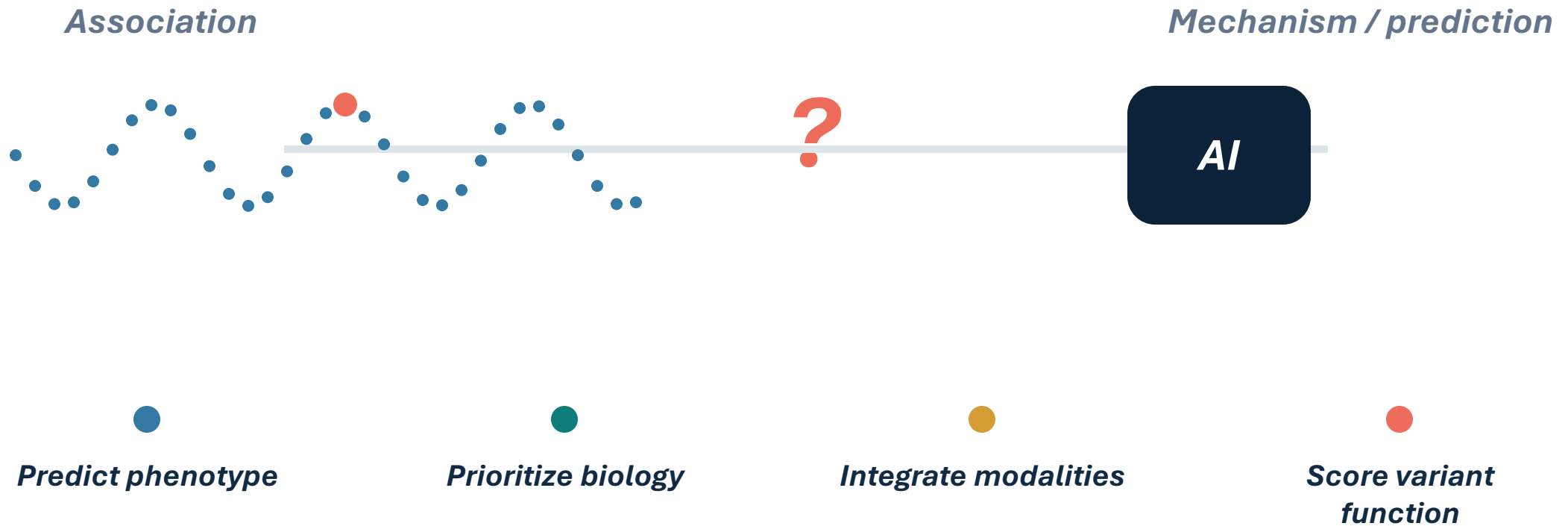


AI revolution *everywhere*

- *Detect complex pattern*
- *Analyse large datasets*
- *Integrate heterogeneous source of information*



A GWAS locus is a beginning, not an answer

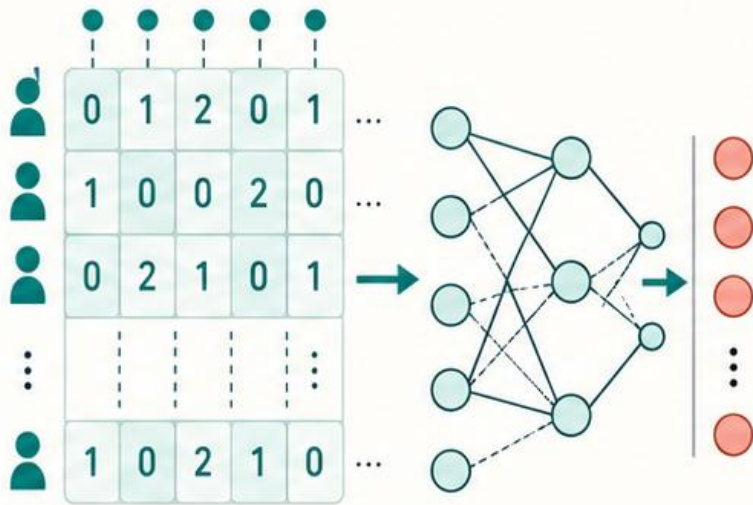


The model is useful only if its output answers the scientific question.

AI in genomics is not one problem

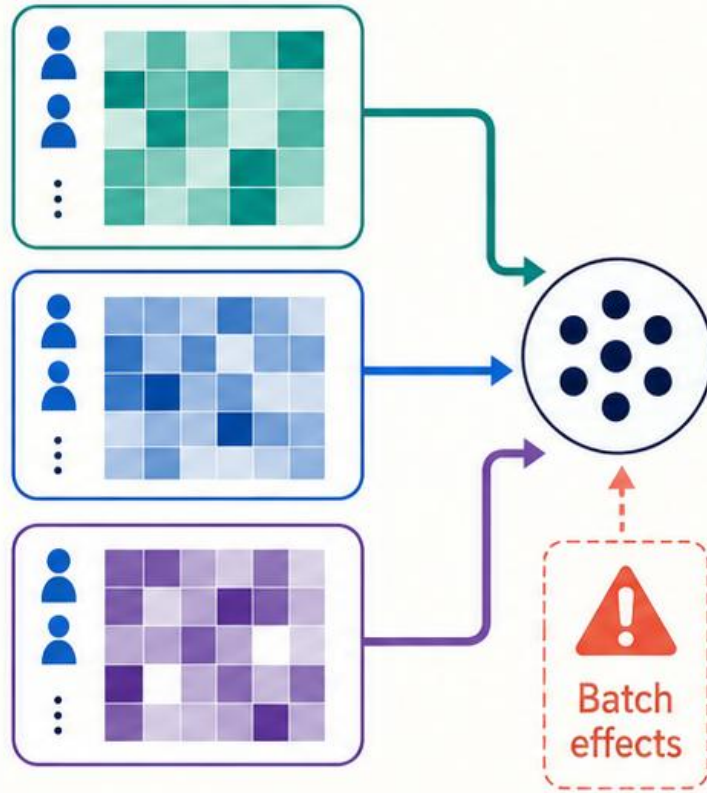
People × variants

Predict phenotypes



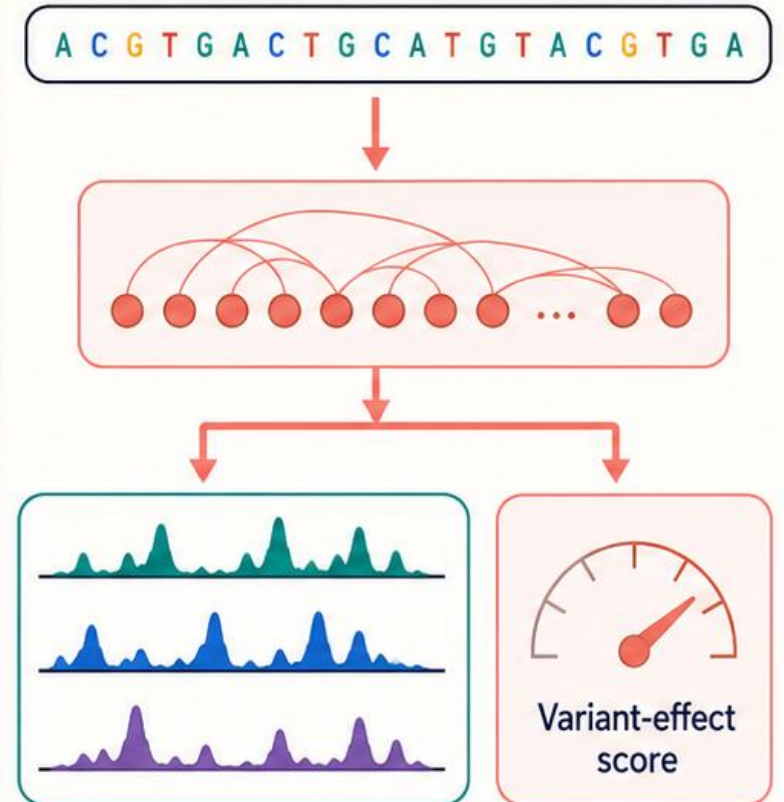
People × omics

Integrate modalities



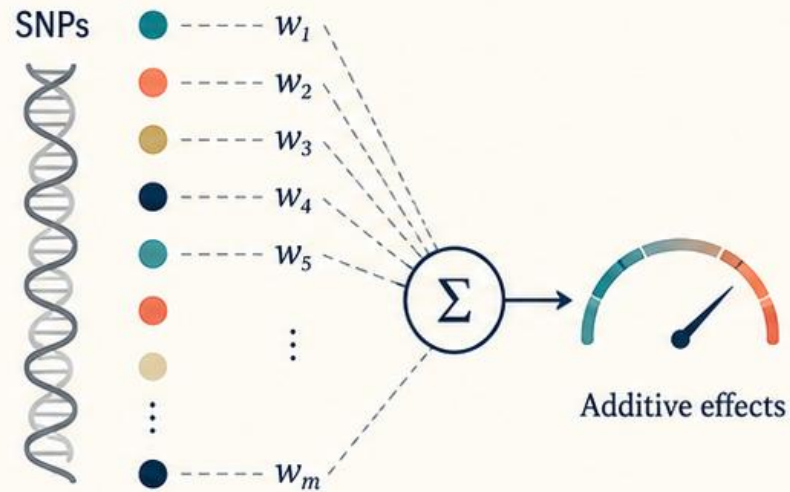
DNA sequence

Predict function



From polygenic risk scores to machine learning

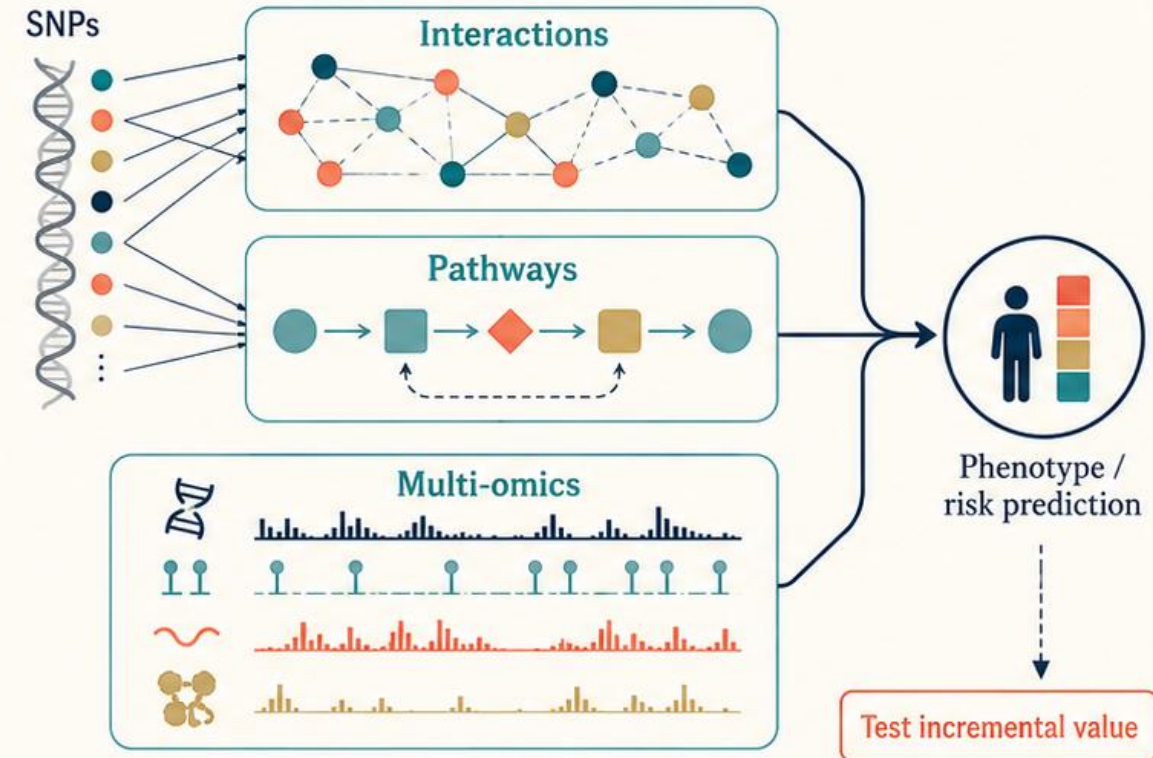
POLYGENIC RISK SCORE



What additional structure can we learn?

Strong baseline

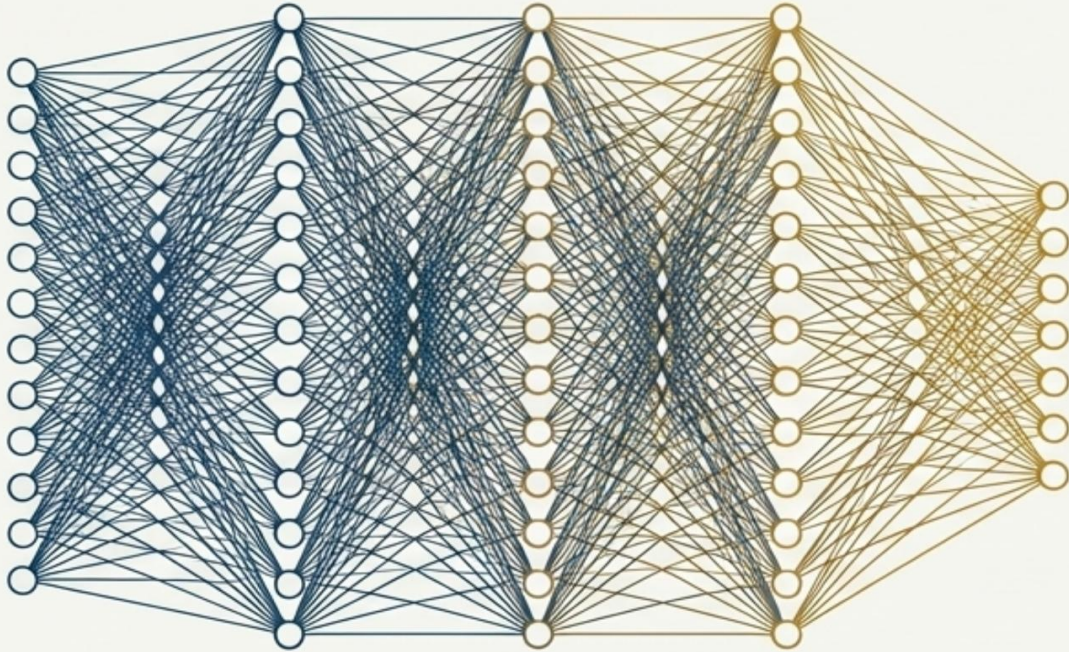
MACHINE LEARNING



Genotype → phenotype

When millions of variants meet thousands—not millions—of participants.

The Deep Learning Dilemma in Genomics



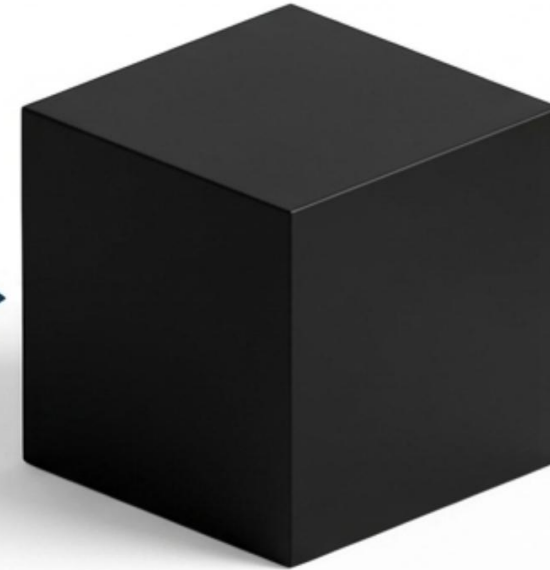
Problem 1: The 'Black Box' Issue

Powerful algorithms like fully connected neural networks are often uninterpretable due to their complexity. They provide predictions without explanations.

Problem 2: The Computational Wall

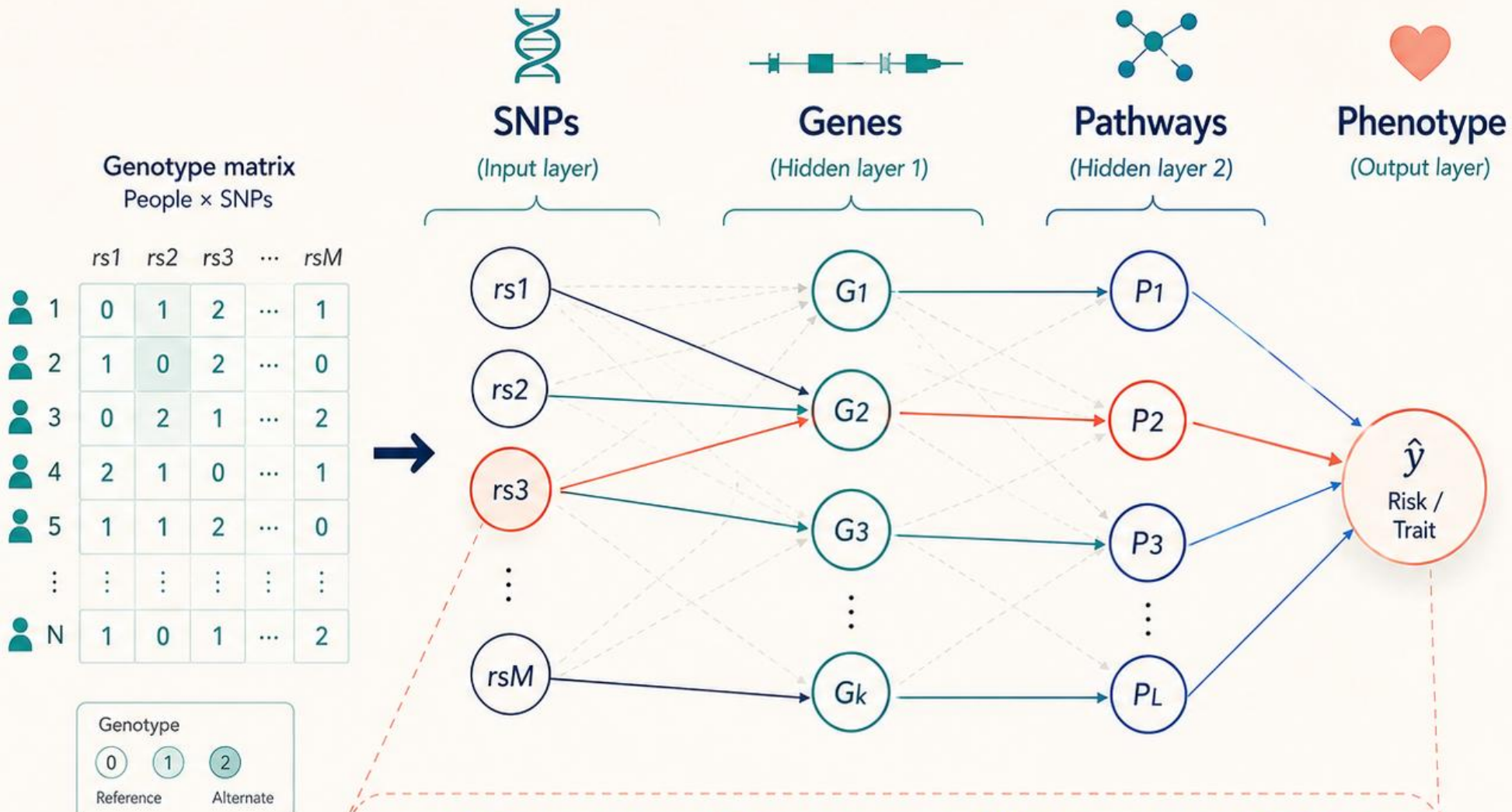
These networks are extremely memory-intensive. Applying them to millions of input variants is often computationally infeasible, requiring pre-selection of SNPs via GWAS, which undermines the unbiased approach.

Genetic Data

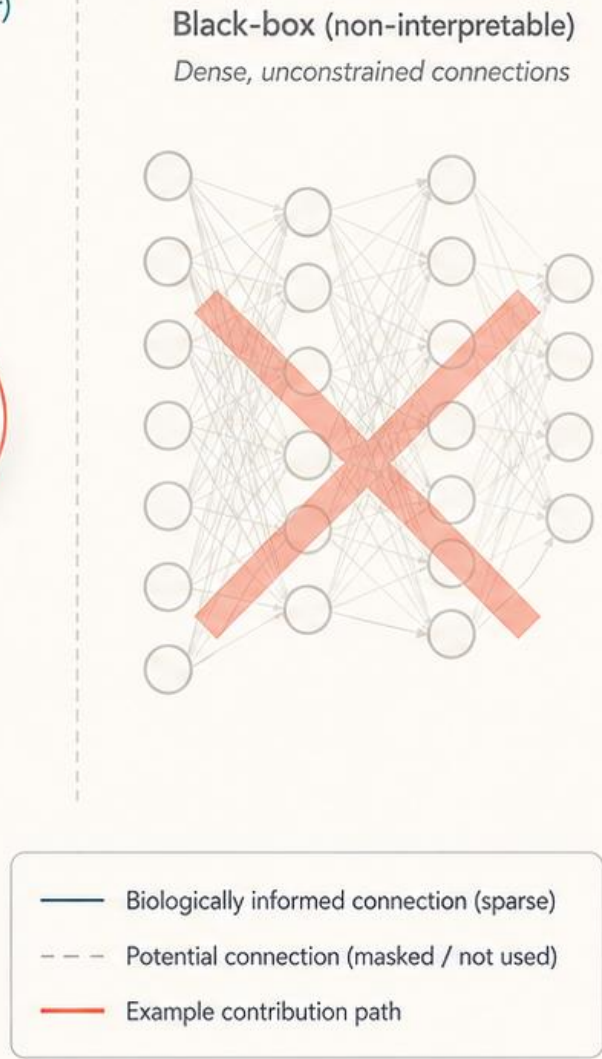
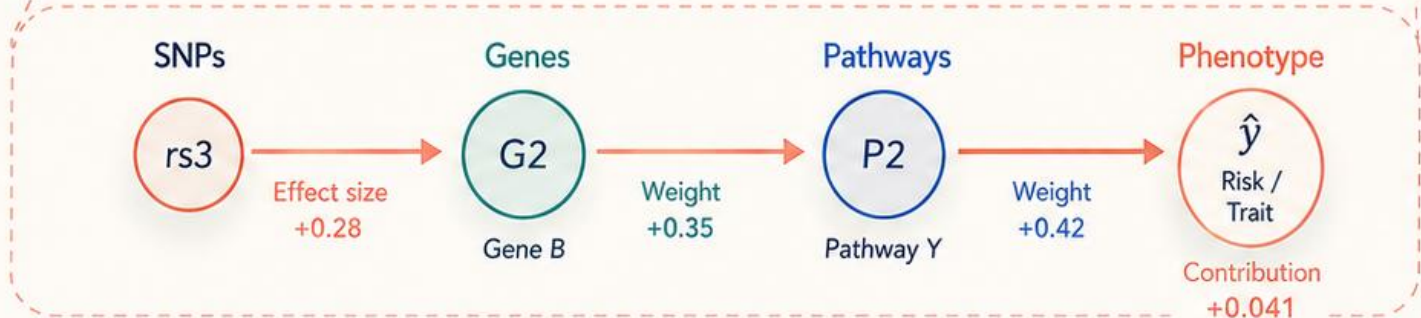


Prediction

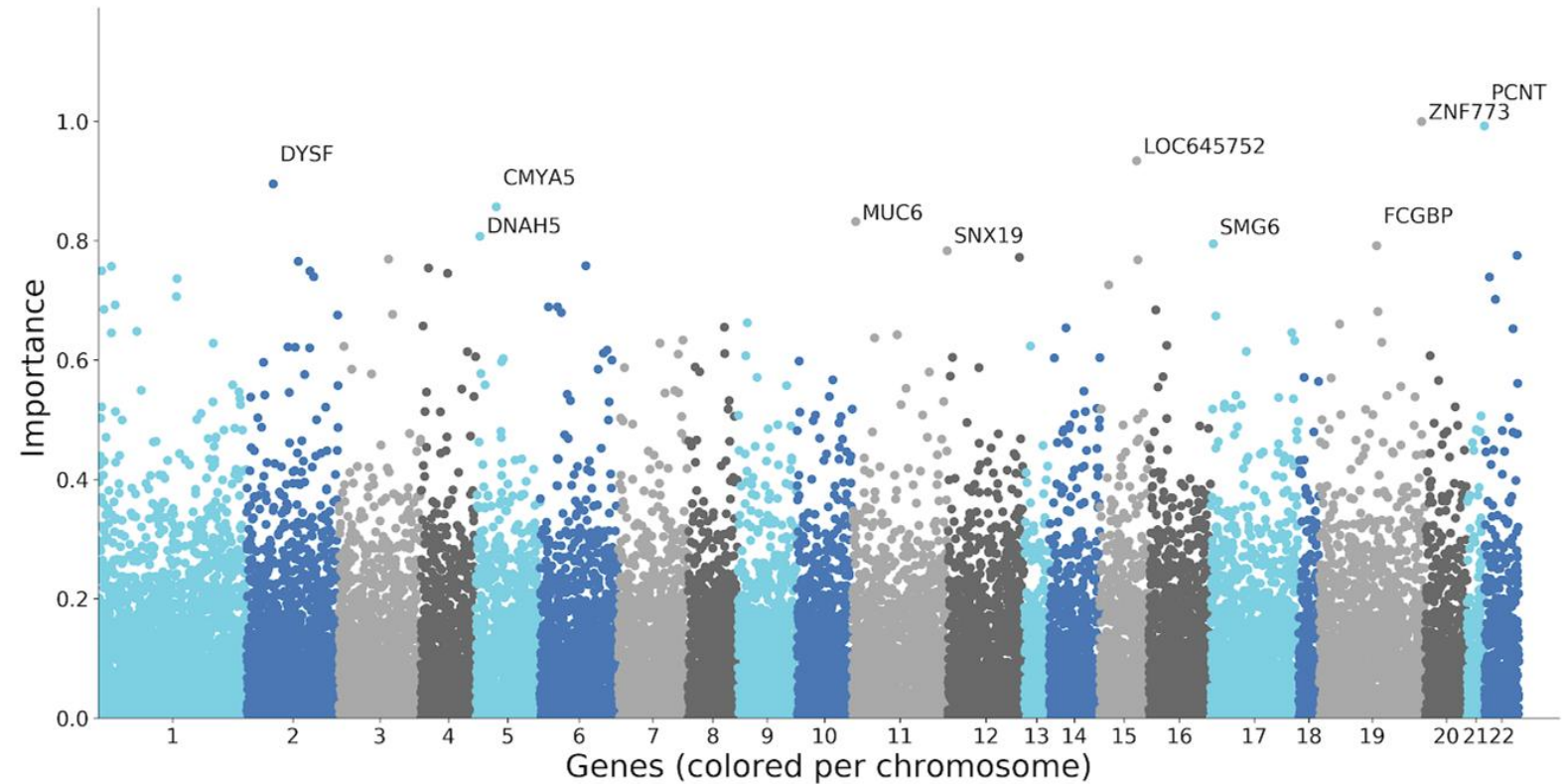
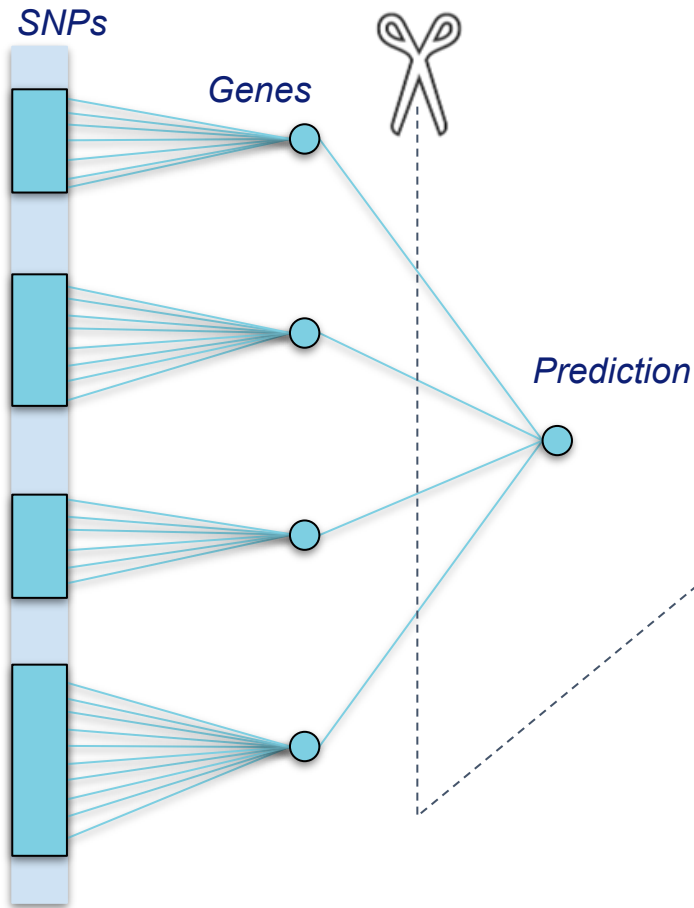




How one variant contributes to the phenotype

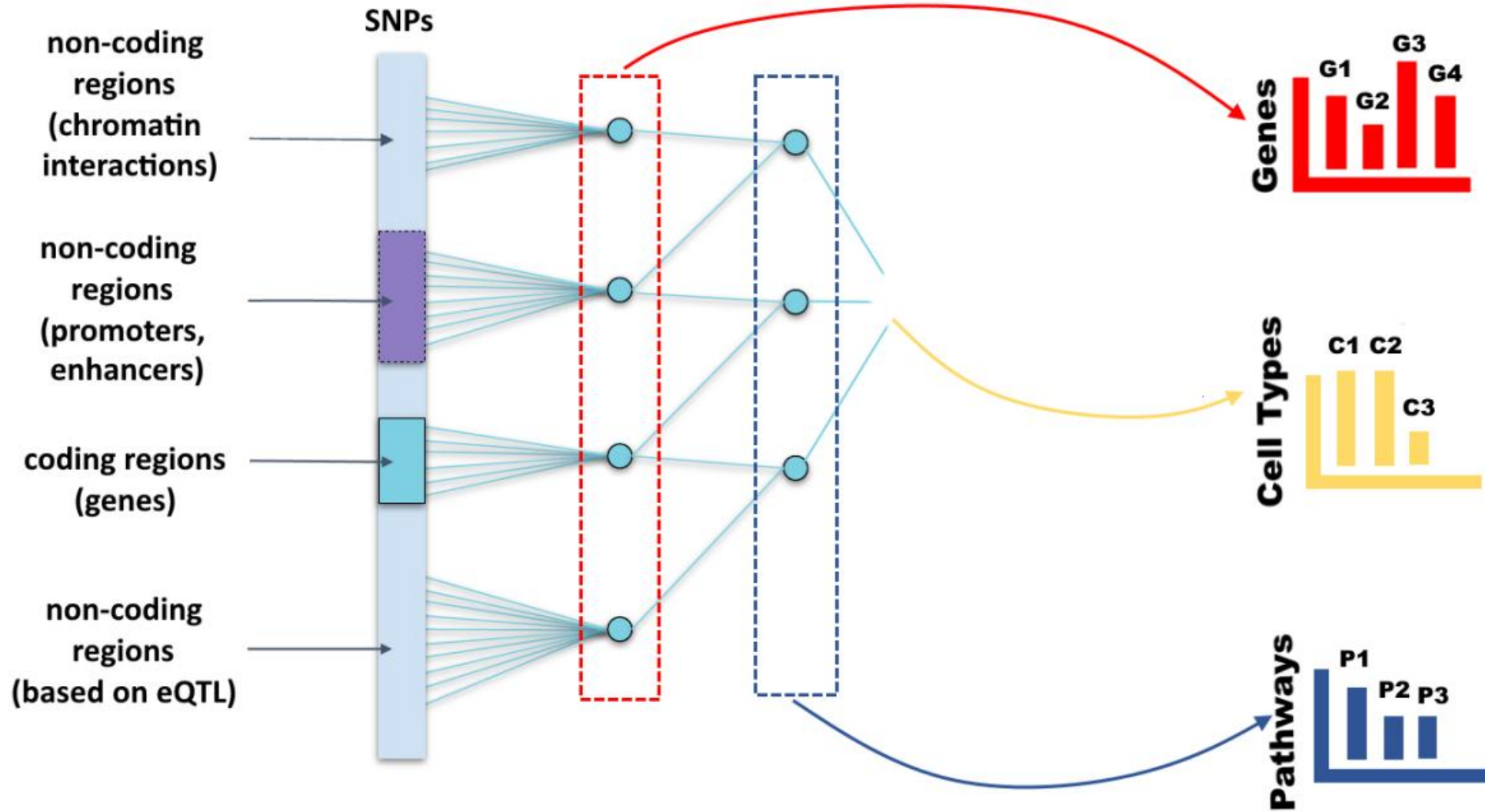


Interpretable Neural Network: GenNet



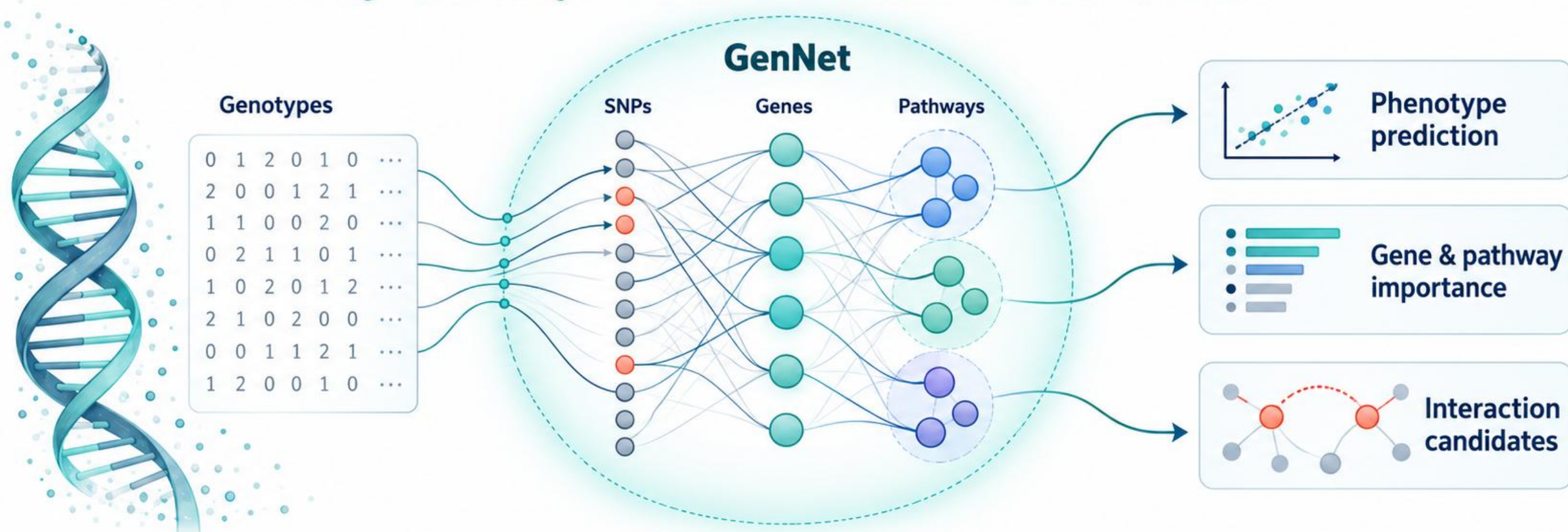
van Hilten, A., Kushner, S.A., Kayser, M., Ikram, M.A., Adams, H.H., Klaver, C.C., Niessen, W.J. and Roshchupkin, G.V., 2021. GenNet framework: interpretable deep learning for predicting phenotypes from genetic data. *Communications biology*, 4(1), p.1094.

Interpretable Neural Networks



ALIEN: Artificial Intelligence Explains geNetics

Biological knowledge • Visible neural networks • Genetic interactions



ALIEN research programme

Research directions



Configurable
biological priors



Cross-cohort
learning



Simulation
benchmarks

Explore AI & Machine Learning in Healthcare

SCAN TO EXPLORE



Interactive ML course

Start where you are.
Learn at your own pace.



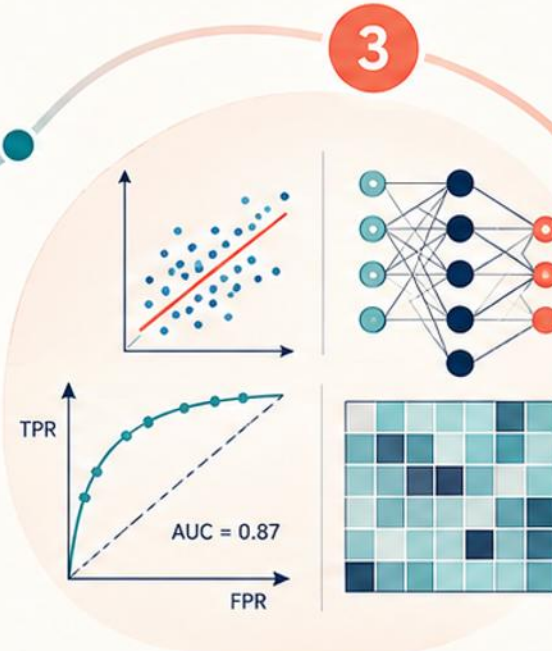
CHECK YOUR LEVEL

65+ question interactive quiz



FOLLOW YOUR ROADMAP

Beginner → Intermediate → Advanced



LEARN BY EXPLORING

Interactive visuals, animations & tools



FROM FOUNDATIONS TO PRACTICE

Algorithms • Validation • Neural Networks
• Fairness • Clinical AI



www.roshchupkin.org/ml-health-slides/



Prof. Gennady Roshchupkin

Your GenNet practical

Neurepiomics 2026

Simulated genotypes → prediction → interpretation

Open the workshop



Scan → click “Open in Colab”

[github.com/roshchupkin/
gennet-neurepiomics-workshop-2026](https://github.com/roshchupkin/gennet-neurepiomics-workshop-2026)

Runs on CPU

1



Simulate a cohort

1,600 people · 192 SNPs
Known effects and an interaction

2



Inspect the biology

Follow the connections:
SNPs → genes → pathways

3



Train GenNet

Fit a small model
on the simulated cohort

4



Compare prediction

Held-out AUC:
GenNet versus lasso

5



Rank genes and SNPs

Inspect gene importance
and the SNP weight plot

6



Check interactions

Rank NID candidates
Check an additive-only control

Can we predict the phenotype — and recover the signals we planted?

A teaching-scale GenNet model within the broader ALIEN research programme

Thank you!

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Department of Epidemiology

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